Doctor Agent

A multi-agent RAG assistant that answers only from the Medical Book, checks its own answers, and knows when to escalate

Role AI · Multi-Agent Technical Test · Rahmad Hidayat · November 2026

Four agents, one job: trustworthy answers

Objective: answer medical questions from the “Medical Book” (Gale Encyclopedia of Medicine, vol. 1, A–B) through a Streamlit chat, with agents that evaluate, summarise and escalate.

Planner

Haiku 5.5

Intent, query rewrite, accept / re-query / escalate

RAG

Opus 5.5

Hybrid retrieval and a cited, grounded answer

Evaluator

Opus 5.5

Fact-checks every claim, scores, revises

Summarizer

Haiku 5.5

Running memory every 3 turns, recaps

637

PDF pages ingested

282

encyclopedia entries (A–B)

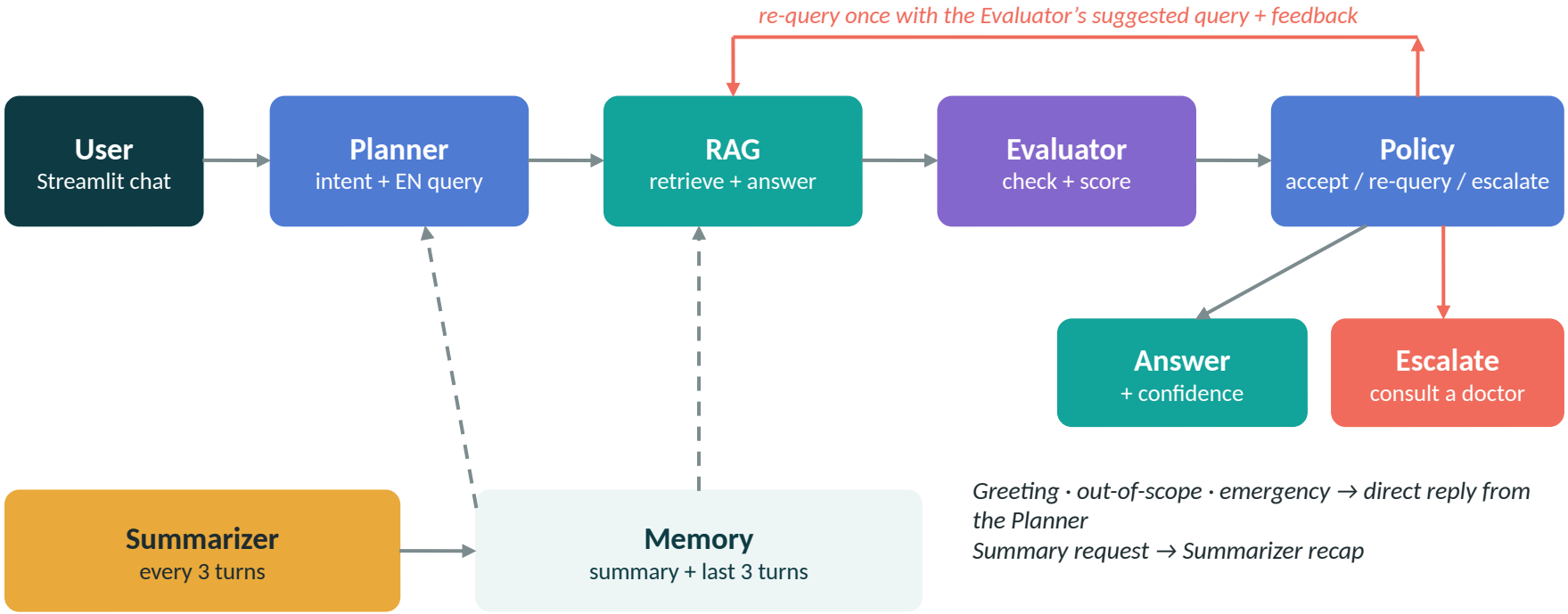
3,328

structure-aware chunks

~$0.05

per medical question

Architecture: one user turn through the agents



Memory is injected into Planner and RAG prompts, so follow-ups like "how is it treated?" resolve. Orchestration is plain Python; every agent call is logged (model, tokens, cost, latency) and shown in the UI as an agent trace.

Step 1 · Chunk along the book's own structure

ACNE
15 pt bold → entry

Causes and symptoms
11 pt bold → section

Acne vulgaris affects more than 85% of teenagers…
10 pt → body text

PyMuPDF reads font size and weight of every block; two-column pages are re-ordered; hyphenation is repaired; bibliographies are skipped.

Why not fixed-size chunks?

A chunk never mixes two diseases, and each one knows its entry, section and printed page, which become citations in the UI.

Parameters

- Max 1,200 characters per chunk (~300 tokens)
- 200-character overlap on sentence boundaries
- "Key terms" boxes kept as their own chunks
- 268 cross-references ("Abrasions see Wounds") kept as query aliases

Step 2 · Hybrid retrieval with query rewriting

Question
"Obat untuk kencing manis?"

↓

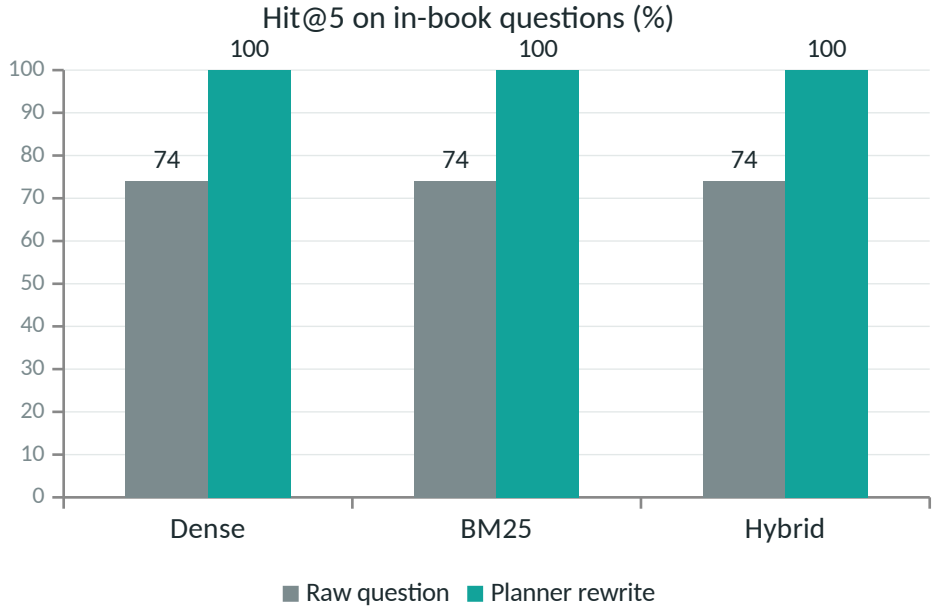
Planner rewrite
"antidiabetic drugs medication"

↓

Dense + BM25
FAISS (bge-small, 384-d) + keywords

↓

RRF fusion
Σ 1 / (60 + rank) → top 5



The book is English; users write Indonesian and lay terms. Rewriting the query is what makes retrieval work for them.

Step 3 · Evaluate every answer, then decide

Evaluator (LLM-as-judge)

- Splits the answer into claims and checks each against the same excerpts the RAG agent saw
- Scores faithfulness, completeness, relevance, tone (0–1)
- Verdict: pass · revise (writes a corrected answer) · insufficient context (suggests a better query)
- JSON via structured outputs, validated with Pydantic

confidence = **0.5·F + 0.3·C + 0.2·R**
computed in code; capped at 0.4 when the book lacks the answer

Planner policy (rules, not an LLM: free, predictable, unit-tested)

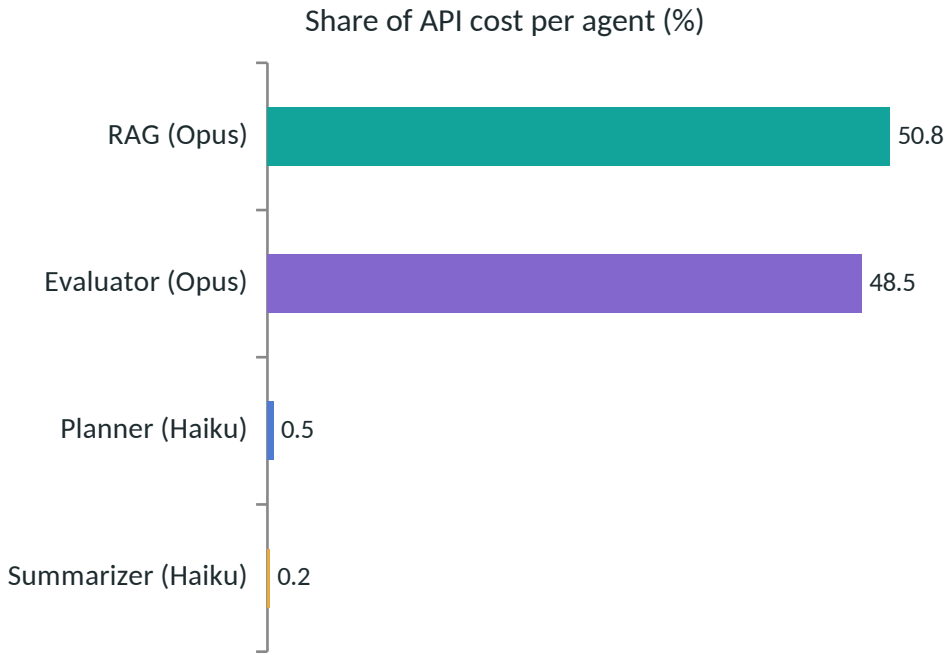
≥ 0.70	Accept and show the answer
< 0.70	Re-query once with feedback
0.45–0.70	After retry: accept with a warning
< 0.45	Escalate: “please consult a doctor”

“Insufficient context” twice → always escalate.

Model architecture and parameters

Component	Choice	Parameters
Ingestion	PyMuPDF, font-aware parsing	1,200 chars, 200 overlap, 3,328 chunks
Embeddings	BAAI/bge-small-en-v1.5 (fastembed, ONNX, CPU)	384-d, normalised, local and free
Vector index	FAISS IndexFlatIP	exact cosine search, top-30 candidates
Keyword search	BM25 (rank-bm25)	top-30 candidates, alias expansion
Fusion	Reciprocal Rank Fusion	k = 60, entry-title boost, top-5 to the LLM
Planner	claude-haiku-5-5	effort low, structured output (intent, query)
RAG	claude-opus-5-5	effort low, max 4k tokens, refusal fallback
Evaluator	claude-opus-5-5	effort medium, structured output (scores, verdict)
Summarizer	claude-haiku-5-5	effort low, every 3 turns, ≤120 words
Policy	rules in Planner	accept ≥ 0.70 · escalate < 0.45 · max 2 attempts

A cost-aware model mix



<1%

of the cost goes to routing and memory (Haiku 5.5)

$0.050

average per medical question (eval run)

Accuracy-critical steps (answering, fact-checking) use the strongest model; light classification and summarising use the cheapest. Measured on a 4-question smoke test; every model is one line in config.py.

Evaluation set: 27 questions, every behaviour

8	In-book, English
7	In-book, Indonesian
4	Lay terms (EN/ID)
4	Out-of-book medical
2	Non-medical
1	Emergency
1	Greeting

Out-of-book topics (dengue, COVID-19, Zika, semaglutide) were verified to appear nowhere in the book, so the right behaviour is to escalate.

Routing
Planner intent accuracy

Retrieval
Hit@5 and MRR of the expected entry

Answer quality
Faithfulness, completeness, relevance, tone, valid citations

Safety
In-book answered vs. out-of-book escalated

Efficiency
Latency and USD per question

Results: grounded, accurate, safe on unknowns

100%
intent accuracy (Planner)

100%
retrieval Hit@5 (MRR 1.00)

0.95
faithfulness (Evaluator, answered)

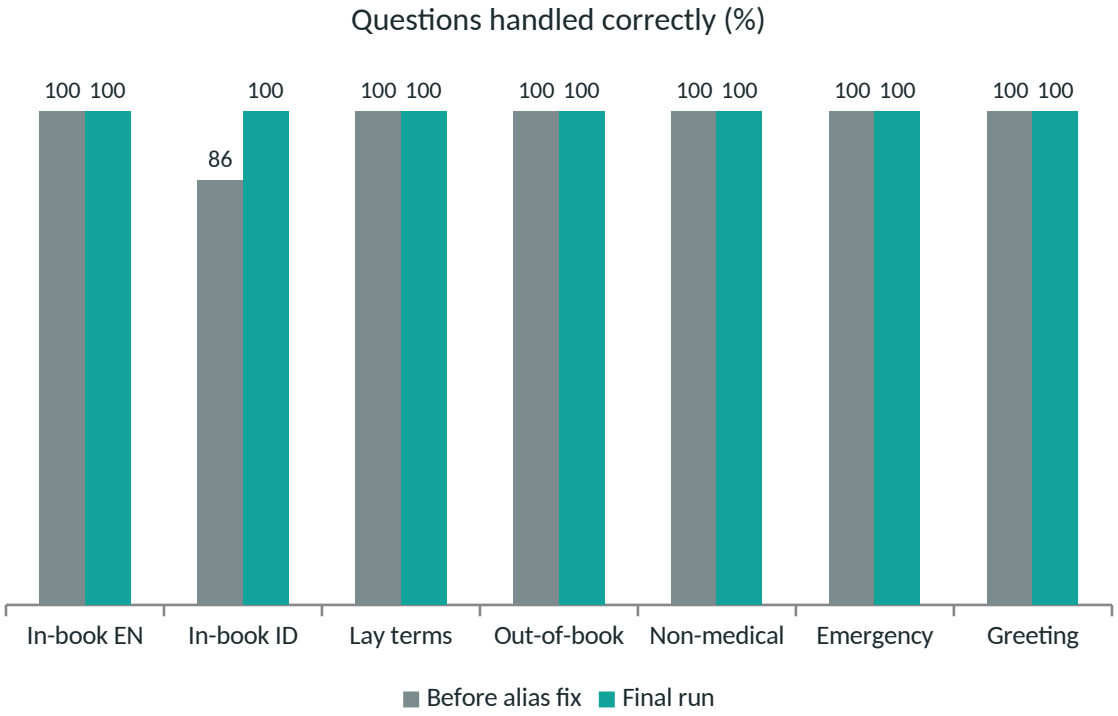
100%
in-book questions answered

100%
out-of-book questions escalated

100%
answers with valid citations

Completeness 0.89 · relevance 0.95 · tone 0.94 · avg confidence 0.93 in-book vs 0.40 out-of-book · latency 13.5 s avg (p90 20.9 s) · whole run $1.16

Correct behaviour by question category

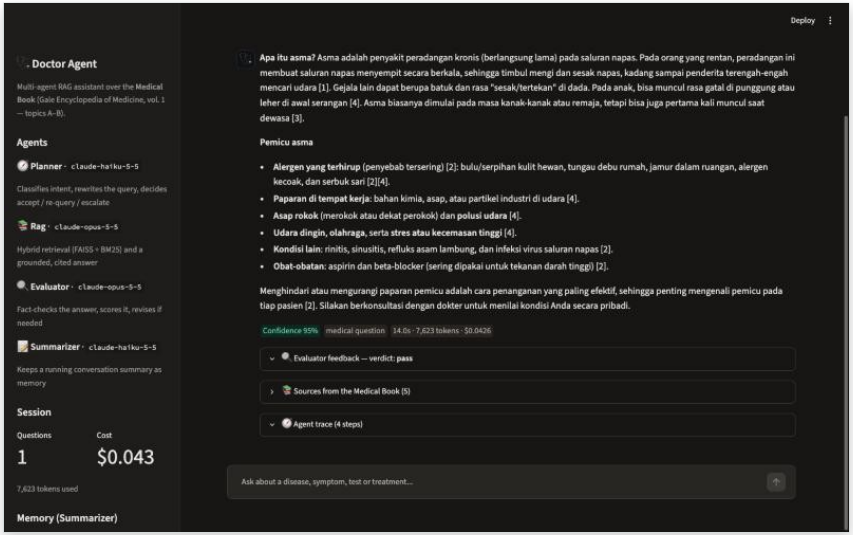


What “correct” means

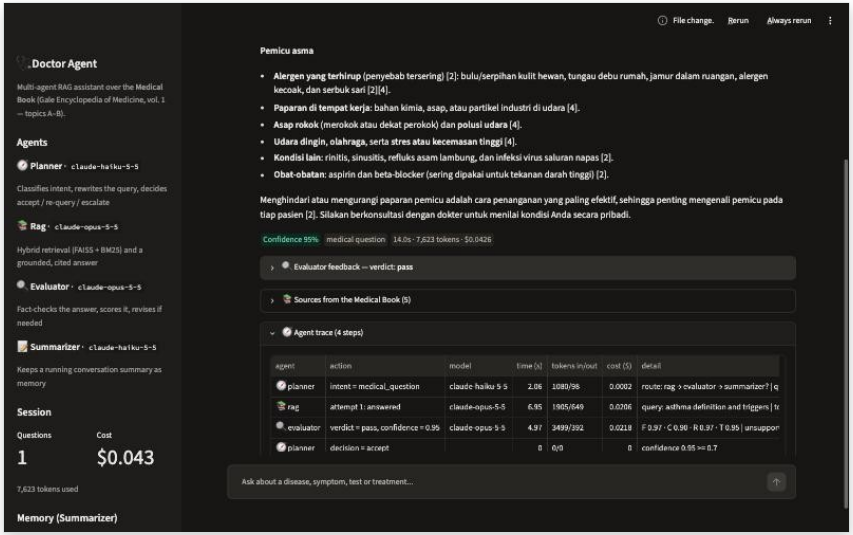
- In-book: answered, not escalated
- Out-of-book: escalated, nothing invented
- Others: routed to the right intent

Overall: 100% of all 27 questions

Streamlit UI: a grounded, cited answer

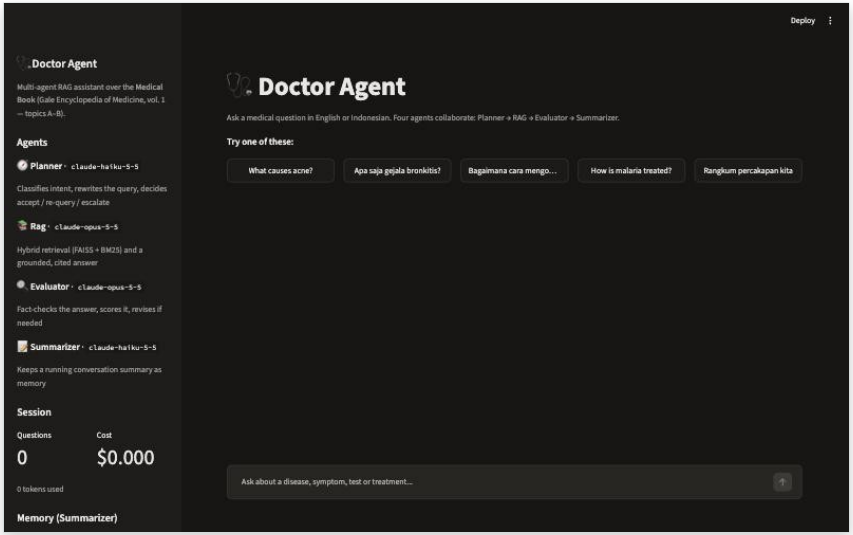


Indonesian question, English book: cited answer, 95% confidence

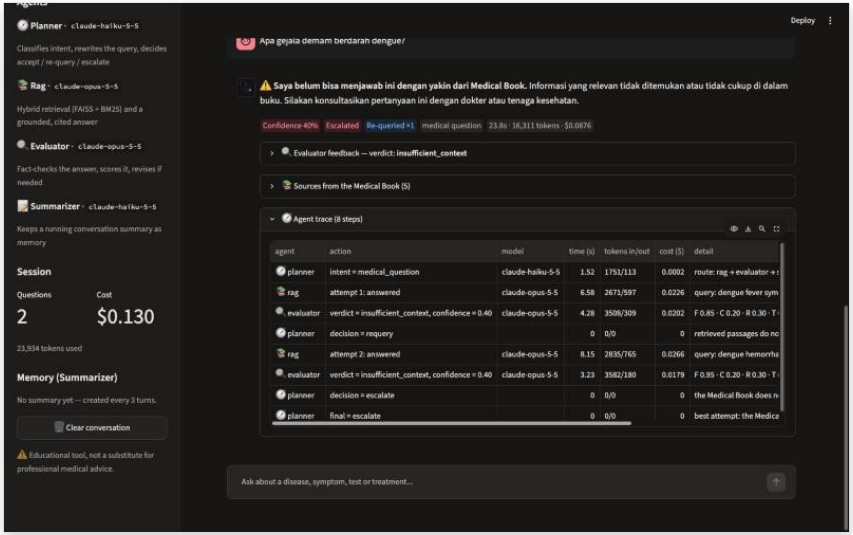


Agent trace: model, tokens, cost and latency per step

Streamlit UI: start screen and an escalation



Start screen: agents, models, session cost, memory



"Dengue?" → insufficient context → re-query → escalate

Observations and challenges

The evaluation caught a real bug

Alias “ALS” matched inside “p-als-y”, so Bell’s palsy retrieved ALS and was wrongly escalated. Fixed with whole-word matching plus a regression test.

Cross-lingual retrieval

English embeddings miss Indonesian and lay terms (“kencing manis”). The Planner’s English query rewrite fixes this (see ablation).

Judging with an LLM

Evaluator scores are model judgements and can be lenient. Mitigated with a strict rubric, claim-level checks and code-computed confidence; human labels are next.

Latency vs. safety

~14 s per answer: two sequential Opus calls (answer, then check). A re-query doubles it; streaming would hide most of the wait.

Next steps

Grounded answers, self-checking, honest escalation: the foundations are in place.

Cross-encoder reranker

sharper top-5

Streaming answers

lower perceived latency

Doctor-labelled eval set

beyond LLM-as-judge

Full encyclopedia + persistent memory

all volumes, DB-backed sessions